

Performance Marketing Analysis — Case Study

Personal Data Analytics Project | Public marketing campaign dataset (200,000 campaigns, 2021) | Tool: Power BI | Role: Data Analyst (self-directed)

1. Project Overview

This self-directed project analyses a public dataset of 200,000 digital marketing campaigns run across six channels (Google Ads, Facebook, YouTube, Instagram, Email and Website). The goal was to build a Power BI dashboard that shows how campaigns perform on reach, conversion, return on investment (ROI) and audience engagement, and to test whether any channel, campaign type, customer segment or spending level clearly outperforms the others.

2. Business Questions

- Which channels and campaign types deliver the best ROI and conversion rate?
- Which customer segments are the most engaged?
- Does spending more per campaign lead to a higher ROI?
- How does click volume change through the year?

3. Dataset

One row per campaign, with campaign type, channel, target audience, customer segment, duration, date, location, language, acquisition cost, clicks, impressions, conversion rate, ROI and engagement score (1 to 10). Data source: public marketing campaign dataset; no employer or client data is used.

4. Data Preparation and Checks

Item	What I found	What I did
Acquisition_Cost	Stored as text with \$ and commas	Cleaned and converted to a number so it can be summed and averaged
Duration	Text such as "30 days"	Split off the number and stored it as whole days (15, 30, 45, 60)
Date	Month/day/year text	Converted with the US locale so days and months are not swapped
Conversion_Rate	Checked in Power Query	100% valid; minimum 0.01, maximum 0.15, so values are proper proportions
Row count	200,000 campaigns	Confirmed with a campaign-count measure after loading

5. Metrics and Measures

Metric	Definition
CTR	Total clicks ÷ total impressions

Metric	Definition
CPC	Total acquisition cost ÷ total clicks
Avg ROI	Mean of the ROI column
Avg Conversion Rate	Mean of the conversion rate column (rates are averaged, never summed)
Avg Engagement	Mean engagement score (scale 1 to 10)

6. Quality Checks on the Dashboard

The first version of the dashboard contained calculation errors, so every chart was re-checked before the final version. Each chart now passes three tests: the right aggregation is used (averages for rates and scores); group values centre on the overall KPI; and values fall inside the possible range.

Issue in the first version	Fix
Conversion chart showed bars of 146%, 122% and so on	Bars were a share of the grand total. Replaced with the average conversion rate per campaign type (about 8%)
ROI by channel used a custom weighted measure that created gaps (7.0 vs 3.1)	Replaced with the plain average ROI; all channels sit near 5.0
Cost vs ROI plotted sums at company level	Replaced with average ROI by acquisition cost group
Dashboard was filtered to a single month	Filter cleared so all 200,000 campaigns are included

7. The Dashboard

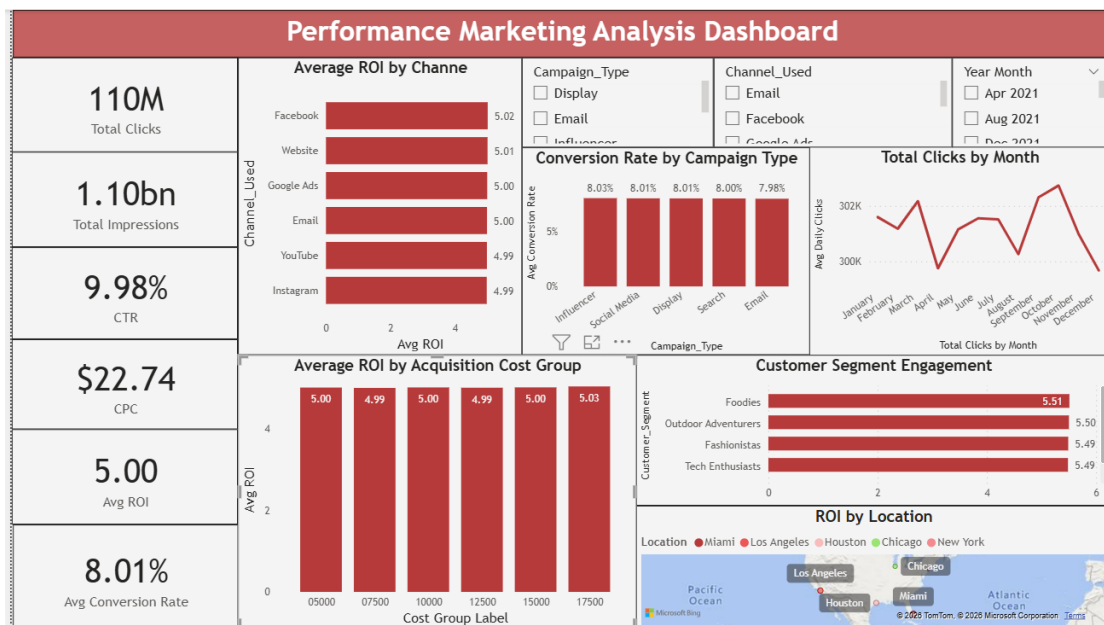


Fig. 1 — KPIs, average ROI by channel, conversion rate by campaign type, average daily clicks by month, average ROI by acquisition cost group, customer segment engagement and ROI by location.

110M	1.10bn	9.98%	\$22.74	5.00	8.01%
Total clicks	Total impressions	CTR	CPC	Avg ROI	Avg conversion rate

8. Key Insights

- **ROI is almost identical across channels.** Facebook 5.02, Website 5.01, Google Ads 5.00, Email 5.00, YouTube 4.99, Instagram 4.99. The overall average is 5.00, so channel alone does not explain return.
- **Conversion rate is almost identical across campaign types.** Influencer 8.03%, Social Media 8.01%, Display 8.01%, Search 8.00%, Email 7.98%, against an overall 8.01%.
- **Engagement is the same in every customer segment.** Scores are about 5.5 out of 10 in each segment, and segments differ by only a few hundredths of a point.
- **Spending more per campaign does not raise ROI.** Average ROI stays between 4.99 and 5.03 for every acquisition cost group from \$5,000 up to \$17,500 and above.
- **Click volume is steady.** Average daily clicks stay at about 300K across the year, with no strong seasonal pattern.

9. Recommendations

- Do not move budget between channels or campaign types on this evidence alone; the data shows no clear winner.
- Run controlled A/B tests on creative, audience and landing pages before changing spend.
- Do not assume a larger acquisition budget will improve returns; check cost per conversion and lead quality alongside ROI.
- On real company data, repeat this analysis by campaign and test whether any differences are statistically significant before acting.

10. Data and Limitations

- Public sample dataset. The very even results across all groups suggest it may be synthetic, so small differences should not be treated as real effects.
- Conversions are not counted directly in the data, and CPC here is total acquisition cost divided by total clicks.
- The results show association only; they do not show cause.
- Only five cities are present, so the location view is descriptive and supports no geographic conclusion.

11. Skills Applied

Power BI, DAX, Power Query, data cleaning, KPI design, dashboard design, validating results, marketing analytics, data storytelling.

12. Conclusion

This project took a large campaign-level dataset from raw text fields to a validated dashboard. Its main lesson is analytical discipline: checking each chart against the overall numbers exposed calculation errors, and the corrected result is an honest one. In this data no channel, campaign type, segment or cost level stands out, so the sound recommendation is to test before shifting budget.